Instructions for Semester-Long Machine Learning Project
CSC/DSCI 4850/6850: Machine Learning
Summer 2026

1. Project Objective

The goal of this project is to apply machine learning techniques to a real-world or synthetic dataset, gain hands-on experience with end-to-end ML workflows, and present meaningful insights or results.

2. Group Formation

- Work in groups of **2 to 3 students**.
- Keep at least 1 graduate and 1 undergraduate students in each team
- If an undergraduate doesn't find any graduate student, a team can be formed with 3 undergraduate students.
- No two graduate students in the same group.
- Ensure a mix of skills within your group (e.g., coding, data handling, and presentation skills).

3. Project Guidelines

3.1. Topic Selection

- Choose a project topic relevant to machine learning, such as:
    - Classification (e.g., image recognition, disease diagnosis)
    - Regression (e.g., predicting house prices, weather forecasting)
    - Clustering (e.g., customer segmentation)
    - Natural Language Processing (e.g., sentiment analysis)
    - Time Series Forecasting (e.g., stock price prediction, energy consumption)
    - Reinforcement Learning (if ambitious).
- Avoid overly simplistic projects (e.g., linear regression on basic datasets).
- Submit your team information and a **brief proposal (as described in the announcement)** describing your idea by **July 1st, 2026,** for approval. Only one submission per team should be fine at the proposal stage. Bul clearly mention team members' name and emails.

3.2. Dataset Requirements

- Use publicly available datasets (e.g., Kaggle, UCI Machine Learning Repository, government datasets) or create your own.
- Ensure your dataset is large enough to provide meaningful insights but manageable for semester work.

3.3. Project Milestones

1. **Proposal Submission**: Define your project goal, dataset, and methods.

2. **Data Preprocessing**: Clean, preprocess, and explore the dataset.
3. **Model Selection**: Experiment with various algorithms and select the best-performing one(s).
4. **Evaluation**: Use appropriate metrics to assess model performance (e.g., accuracy, F1-score, AUC).
5. **Final Report and Presentation**: Summarize your approach, findings, and key takeaways.

3.4. Deliverables

1. **Proposal**: A short document outlining your project plan. Include a timeline of actions. (due by **July 1st**). [**5 points**]
2. **Codebase**: Submit all scripts, notebooks, and related files via iCollege.
3. **Mid-Semester Progress Update**: Provide a brief update on your progress and challenges faced. (**Due by July 15**) [**5 points**]
4. **Final Report**:
    - Follow [IEEE Conference Template]()
    - Include peer evaluation before the abstract and after the title (confidential). Guidance of how to do it will be given later.

    A detailed report (5-7 pages) covering:

    - Introduction and problem statement
    - Dataset description
    - Methods and models used
    - Results and evaluation metrics
    - Discussion and conclusions
    - Future Directions
    - Each member needs to submit separate copies to iCollege dropbox.

    Final report is due by **July 30.** [**10 points**]

5. **Presentation**: Prepare a 15–20-minute presentation with visuals (slides, plots) to present your findings to the class. Graduate students are exempt of project presentations. They will be assigned separate paper presentation tasks. [**10 points**]

4. Additional Notes

- Use tools and libraries like Python, scikit-learn, PyTorch, pandas, and matplotlib.
- Maintain a clear project timeline and distribute tasks evenly among team members.
- Seek feedback during the semester to stay on track.
- Reach out for guidance if you encounter roadblocks.

5. Resources for Inspiration

- Kaggle Competitions: www.kaggle.com/competitions

- UCI ML Repository: https://archive.ics.uci.edu/ml/index.php
- Google Dataset Search: datasetsearch.research.google.com

Relevant Conferences to Explore:

- ICML
- NeurIPS
- ICLR
- CVPR
- ICCV
- UAI
- AISTATS
- KDD
- AAAI
- FAccT

NB: This is just a list of major CS conferences. You don't have to choose papers only from these venues. I understand this class has many undergraduate students, and they don't have any research experience. Choose papers from any Journals/Conferences that are valid.
